A Basis for the 2-Kernel of Euler's Totient

Exact finite-level ranks, a forced carry bound, and where Erdős #249 stops

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Abstract

Erdős #249 asks whether $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational. It is open, nothing here closes it, and this note reports what a formalised attack does yield, each claim bounded by the exact proposition a proof checker accepted — the negative results included.

The 2-*kernel* of a sequence c is the family of subsequences $n \mapsto c(2^j n + r)$ for $j \geq 0$ and $0 \leq r < 2^j$. Coons proved that φ is not k-regular for any $k \geq 2$, that is, that the $\mathbb{Q}$-span of its k-kernel is not finitely generated; Bell and Smertnig strengthened this to non-Mahlerity. Both are statements about the union over all levels. Our main theorem is the complementary positive one, at each finite level: writing $\varphi_{j,r} \colon n \mapsto \varphi(2^j n + r)$, the sections $\varphi_{0,0}$, $\varphi_{1,0}$ and $\varphi_{j,r}$ with r odd form a $\mathbb{Q}$-basis of the span of *all* the $\varphi_{j,r}$; the span through level e has dimension exactly $2^e + 1$; and the only $\mathbb{Q}$-linear relations among dyadic sections of φ are those generated by two elementary identities.

Transporting that rank through binary long division gives the one consequence for S we can state: rationality would force a tempered integral carry whose sections through level e span at least $2^e - 1$ dimensions, for every e. This is an anti-compression barrier and not a criterion, because no finite-rank upper bound is proved for the carries rationality supplies.

We also record the exact reformulations, three equivalences and one implication, kept separate because an equivalence is not progress; a Farey denominator exclusion below 7.96×10^{34}, whose method is classical and whose scale is a function of how many binary digits were committed; and the proof that no fixed positive integer clears the coordinates $A(n)/n$ of the Lambert weight $A = \varphi * \mu$.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Authorship and AI use. The prose in this version was generated by large-language-model agents under Will Cook's direction. Cook set the objectives and acceptance criteria, reviewed and authorised the manuscript, and is responsible for its contents. The agents are production tools, not authors. See *Statements and declarations*.

1 Results

Every item is a proposition the pinned Lean kernel accepted, or (R10) the problem itself. Each *Checked* line links the declarations at commit `571ec44f2aad`. The tag gives the status recorded for that result in the claim registry's taxonomy — and where the registry owns no row for the cited declarations, the tag says so rather than guessing one. Nothing in the list is an irrationality criterion for S, and no combination of the items becomes one.

Throughout $\varphi_{j,r}(n) = \varphi(2^j n + r)$, for $j \geq 0$ and $0 \leq r < 2^j$, is the (j, r) *dyadic section* of Euler's totient, a vector in $\mathbb{Q}^{\mathbb{N}}$; the family of all of them is the 2-kernel $\mathcal{K}_2(\varphi)$ in the sense of Allouche and Shallit [1]. Two elementary identities relate them: the *reductions* $\varphi_{j+1,0} = 2^j \varphi_{1,0}$ (checked) and $\varphi_{j,2^{t+1}s} = 2^t \varphi_{j-t-1,s}$ for s odd with $2^{t+1}s < 2^j$ (checked).

R1. [CHECKED; NO REGISTRY ROW] **Basis theorem.** The family $\mathcal{B} = \{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1,\ r \text{ odd},\ 0 < r < 2^j\}$ is linearly independent over $\mathbb{Q}$ and spans the same $\mathbb{Q}$-subspace of $\mathbb{Q}^{\mathbb{N}}$ as $\mathcal{K}_2(\varphi)$; so $\mathcal{B}$ is a basis of that span, which is therefore infinite-dimensional. Since the reductions make every remaining section a rational multiple of a member of $\mathcal{B}$, the $\mathbb{Q}$-linear relations among dyadic sections of φ are exactly those the reductions generate — a one-line consequence of the checked statements, not a further one. *Checked:* independence, span equality, basis. *Registry:* none of these three declarations is currently owned by a row of the claim registry; see Section 8.2.

R2. [UNCONDITIONAL PROGRESS] **Exact finite-level rank.** For every $e \geq 0$ the $2^e + 1$ sections $\varphi_{0,0}$, $\varphi_{1,0}$ and $\varphi_{j,r}$ with $1 \leq j \leq e$, r odd, $0 < r < 2^j$ are linearly independent over $\mathbb{Q}$, so their span has dimension exactly $2^e + 1 = 2 + \sum_{j=1}^{e} 2^{j-1}$: exact, not an estimate, because the even residues are already dependent by the reductions. R1 follows by finite character of linear independence. *Checked:* independence, dimension, infinite-dimensionality directly.

R3. [FORMALISED HERE] **Denominator exclusion, sharp for its window.** If $S = a/q$ with $a \in \mathbb{Z}$ and $q \geq 1$ then $q > 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}$. The constant is optimal for the window producing it: the next integer up is the exact first denominator at which that window's certificate fails. The method is Farey (Section 4). *Checked:* exclusion, reduced-denominator form, first failure. *Registry:* the row owns the reduced-denominator form only.

R4. [FORMALISED HERE] **Carry anti-compression.** If S were rational then some tempered integral carry orbit for the coefficients φ, with positive multiplier, would have for *every* e carry sections through level e spanning a space of dimension at least $2^e - 1$. *Closes:* every rationality proof that would exhibit a carry of bounded section rank. *Does not close:* anything else — it is not an irrationality criterion, because no finite-rank upper bound is proved for the carries that rationality supplies. *Checked:* the barrier, rank transport from R2.

R5. [CHECKED; NO REGISTRY ROW] **The #249 weight.** $S = \sum_{d \geq 1} A(d)/(2^d - 1)$ with $A = \varphi * \mu$; here $A \geq 0$, $A(p) = p - 2$, $A(p^k) = \varphi(p^k) - \varphi(p^{k-1})$, and A is unbounded. *Checked:* identity, sign, primes, prime powers, unboundedness. *Registry:* the Lambert-ladder row owns the neighbouring rungs, not these five declarations; see Section 8.2.

R6. [UNCONDITIONAL PROGRESS] **Eventually periodic weights are settled.** For every integer base $b \geq 2$, if $\gamma\colon \mathbb{N} \to \mathbb{Q}$ is eventually periodic, nonnegative and positive at some positive index of its periodic part, then $\sum_a \gamma(a)/(b^a - 1)$ is irrational. Luca and Tachiya prove a strictly broader theorem by other methods — every eventually periodic rational sequence that is not eventually zero, at every integer base with $|q| > 1$ [6] — so no priority is claimed and the nonnegativity hypothesis here is not needed for the conclusion. By R5 the weight A is unbounded, hence not eventually periodic, so #249 is outside this class. *Checked:* the periodic theorem.

R7. [PROVED HERE] **No fixed index clears the normalised weight.** For every integer $D \geq 1$ some $n \geq 1$ has $D \cdot A(n)/n \notin \mathbb{Z}$, and any D clearing every coordinate up to a horizon $N \geq 4$ is divisible by an explicit two-tier primorial (4 at $p = 2$, p^2 when $p^2 \leq N$, else p). *Closes:* every argument clearing the coordinates $A(n)/n$ by a single positive integer valid at all n. *Does not close:* arguments whose index grows with n, or which never clear the coordinates. *Checked:* no fixed index, the primorial divisibility.

R8. [PROVED HERE] **Three exact reformulations, offered as nothing more.** Irrationality of S is equivalent to each of: pointwise non-integrality of every tail difference; a cofinal supply of finite certificates at every fixed shift; and the same supply restricted to the lcm diagonal. A fourth equivalence, independent of the certificate language, is a counted anti-concentration condition on window phases. All four are restatements of #249 in other coordinates, and are the part of this work most likely to be mistaken for a result. *Checked:* pointwise, cofinal at each shift, lcm diagonal, window-separated pairs.

R9. [CONDITIONAL REDUCTION] **One implication, not an equivalence.** A supply of gap certificates — one window (N, K) per denominator q whose totient carry residue avoids a band of width $q(N+K+2)$ out of 2^K — *implies* irrationality. The converse is not proved and is not claimed; this is why the item is separate from R8. *Checked:* the supply implication. *Registry:* no row currently owns this declaration.

R10. [OPEN] **Erdős #249.** Whether S is irrational. Not proved here.

2 Where #249 sits

If $f = g * \mathbf{1}$, that is $f(n) = \sum_{d|n} g(d)$, then interchanging the two sums gives

$$\sum_{n\geq 1} \frac{f(n)}{2^n} = \sum_{d\geq 1} \frac{g(d)}{2^d - 1}. \tag{1}$$

Four classical coefficient sequences have this shape. The table records the constant, the weight $g = f * \mu$, and its status.

f	$\sum f(n)/2^n$	weight $g = f * \mu$	status
τ	$1.6066951\ldots = E$	$g \equiv 1$	Irrational (Erdős 1948); formalised here in the Lambert form $\sum_{d\geq 1}(2^d - 1)^{-1}$ (checked)
ω	$0.5169428\ldots$	$g = \mathbf{1}_{\text{primes}}$	Irrational at base 2: Tao–Teräväinen, Thm. 1.3 [4]
Ω	$0.5895032\ldots$	$g = \mathbf{1}_{\text{prime powers}}$	Asserted *ibid.* by a similar argument, with the details left to the reader; not a proved theorem there
φ	$1.3676308\ldots = S$	$g = A = \varphi * \mu$	**Open** (Erdős #249)

The first three weights are bounded, taking only the values 0 and 1 (for τ the weight is the constant 1). The fourth is unbounded, by R5. That difference is the one that matters for the route in R6, and we do not claim it is the only difference: A also vanishes on

$n \equiv 2 \pmod 4$, while $\mathbf{1}_{\text{primes}}$ is bounded but no more eventually periodic than A is, so the settled ω row is not an instance of R6 either. It was settled by a different method, discussed in Section 8.

The constant is not opaque. It is worth saying what *is* known exactly about S, because a reader could otherwise take "open" to mean "structureless". The neighbouring Lambert rungs are rational and exactly computable — $\sum_{d\geq 1} \mu(d)/(2^d - 1) = \frac{1}{2}$ (checked) and $\sum_{d\geq 1} \varphi(d)/(2^d - 1) = 2$ (checked) — so S sits one Möbius convolution from a machine-checked irrational and one from a rational. The signed squared-Mersenne lens $S = \frac{1}{2} + \sum_{d\geq 1} \mu(d)/(2^d - 1)^2$ (checked) converges fast enough to recompute the decimal above.

There is also an exact probabilistic reading, and it is more than a gloss. Let X and Y be independent fair-coin geometric waiting times on $\mathbb{N}_{>0}$. Then

$$S = \tfrac{1}{2} + \Pr[\gcd(X, Y) = 1]$$

(checked), the probability being the visible-coprime-pair series $\sum_{\gcd(m,n)=1} 2^{-(m+n)}$. That probability is distributed over the Stern–Brocot tree in a completely explicit way: the Mersenne weights of the positive reduced slopes sum to exactly 1 (checked), and at each node (a, b) the cylinder mass $1/((2^a - 1)(2^b - 1))$ splits exactly into its stop mass and the masses of its two children (checked). So $S - \frac{1}{2}$ is the total mass of an exactly self-similar subdivision of a probability space. What is missing is not structure; it is an arithmetic consequence of the structure.

3 The 2-kernel of Euler's totient

3.1 What was already known, and what is different here

For an integer $k \geq 2$, the *k-kernel* of a sequence c is $\{n \mapsto c(k^j n + r) : j \geq 0,\ 0 \leq r < k^j\}$, and c is *k-regular* when the module generated by its k-kernel is finitely generated; both notions are due to Allouche and Shallit ([1], Definitions 1.1 and 2.1). Coons proved that φ is not k-regular for any $k \geq 2$ ([2], Theorem 3.3), and Bell and Smertnig have since strengthened this: the totient generating series is not k-Mahler for any $k \geq 2$ [3].

Both are negative statements about the union over all levels: no finite set of sections generates the rest. R1 and R2 are the complementary positive statements at each finite level, and the two are consistent precisely because the rank is unbounded in e. Knowing that a span is not finitely generated says nothing about its dimension at a given truncation, nor about which relations hold there; what is added here is that the dimension through level e is exactly $2^e + 1$, that an explicit family is a basis, and that the two elementary reductions generate every relation. We found no prior source giving any of those three for φ; that is the outcome of a search, not a proof of non-existence, and no priority is claimed.

3.2 The mechanism

The producer for R2 is a separated evaluation minor. For each level, the Chinese remainder theorem together with Dirichlet's theorem on primes in arithmetic progressions supplies a set of evaluation points at which the section matrix has a nonzero minor with the required two-adic separation: one affine channel is made prime, and every other channel receives a fresh prime divisor congruent to 1 modulo a large power of two, so the columns carry distinct exact powers of two and the determinant cannot vanish (checked). Even residues never enter, because the second reduction has already carried them to odd-core channels (checked).

R1 then needs only two further steps, and both are short. Linear independence is a property of finite subfamilies, so the level-e producer, applied with e above the largest level occurring in a given finite subset, already gives independence of the whole infinite family $\mathcal{B}$. And the two reductions carry every remaining section onto a rational multiple of a member of $\mathcal{B}$, so the spans agree. That the assembly is short is the point: the level-e theorem was always the whole content, and stating its consequence as infinite-dimensionality understated it. The sharp form is a basis.

Remark. R1 and R2 are statements about φ, not about S; they are true, and their proofs are complete, whether or not S is irrational. Their connection to #249 is not thematic but a single proved transport, and it is R4.

The rationality side. Binary long division turns a hypothetical rational value into an integer carry recurrence: for coefficients $c(n) \le n$, and in particular for $c = \varphi$, the series $\sum c(n)/2^n$ is rational exactly when a tempered integer carry orbit exists (checked). Transporting R2 through that recurrence gives R4. The scope line there is worth repeating: this is an anti-compression barrier, and the complementary finite-rank upper bound, which is what would make it a proof, is not available.

4 The denominator bound

R3 is a Farey exclusion, and we say so before saying anything else about it. The argument is the classical mediant one. The Lean lemma commits the first 240 binary digits of the shifted series as a single residue over the totient carry window $(N, K) = (1, 240)$; a rational a/q can equal S only if it falls into a resulting bad interval of width $243/2^{240}$; that interval is bracketed by two explicit unimodular fractions $a_1/b < c_1/d$ with $bc_1 - a_1 d = 1$; and the mediant lemma (checked) forces any rational strictly between unimodular neighbours to have denominator at least $b + d$. The excluded range is therefore $q \le b + d - 1$, which is the printed constant (checked), and the transport to the series is a tail estimate (checked). *The method is standard, and we claim no novelty for it.*

Two things should be read off correctly. First, the scale. A window of K computed binary digits yields a mediant bound of order $2^{K/2}$: the same pipeline gives 2.49×10^{17} at $K = 120$ (checked) and 7.96×10^{34} at $K = 240$. The constant is a function of how many digits were committed, not a measure of how much the problem has moved. Second, the sharpness. Within its window the bound cannot be improved at all: $b + d$ itself fails the certificate, so $b + d - 1$ is exactly the last denominator excluded. A reader who regards an explicit Farey exclusion as routine is not in disagreement with this note. R1, not R3, is where we would ask a sceptical reader to look.

The bound transfers to the coprimality form of Section 2 with the constant halved: the visible-coprime-pair series has no representation a/d with

$d \le 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526$ (checked), that number being $(q_0 - 1)/2$ for the constant q_0 of R3.

5 Barriers, and two we do not state as theorems

R7 is the one method barrier this note states, and its scope lines are given with it in Section 1. It is worth a sentence of mechanism, because the mechanism is the whole of it: take a prime $p > \max(D, 2)$; then $A(p) = p - 2$ by R5, and $p \nmid D(p-2)$, so $D \cdot A(p)/p \notin \mathbb{Z}$. Being one line is a virtue here. The statement is about the weight the abstract already names, it needs no vocabulary this note has not defined, and it survives a hostile reading.

Two further no-go results in the formal corpus appeared as propositions in an earlier version of this note, and are demoted rather than dropped, because the reason is more useful to a reader than the propositions were.

A fixed-precision transport result (checked) says that at every fixed positive precision, each finite compatible valuation-unit word admits a prefix-locked centred completion. Its statement quantifies over abstract pairs of a two-adic valuation and an odd unit; it mentions neither φ nor S, its content is that a congruence class meets any interval of the corresponding length, and it is a restatement of a lemma printed immediately above it. The class of arguments it closes — deriving a contradiction from a local signature read at a precision fixed in advance, along a finite word — is not one anybody has proposed for #249.

A factor-ideal result (checked, with a sparse-anchor companion at checked) exhibits, for each $t \ge 3$ with $H = \operatorname{lcm}(1, \ldots, t)$, a nonzero integer carry whose forcing letters lie in the ideal generated by $\varphi(H)$, which reproduces the true totient differences at $t - 2$ prescribed indices, and which every finite integer shift polynomial carries to a pair with the same coboundary form, the same ideal memberships and an ℓ^1-weight bound. It does close that hypothesis class. Two limits. The witness is a spike — its state is $-\varphi(H)$ at $t - 2$ points and zero elsewhere, so it stays uniformly bounded while the strip bounds it respects grow, and the construction is cheap for that reason. And the shifted pairs are not asserted to be nonzero, so an argument that additionally demanded non-vanishing after the shift is not closed. An earlier draft of this note also attributed to it a whole-ray anchor condition, a diagonal-bound condition and a strict-survivor condition that the cited theorem does not contain, and described it as "surviving" the shift algebra when the checked conclusion carries no non-vanishing clause. Correcting that is the reason it is here at all.

6 Three problem-owned programmes

The modules below are problem-owned rather than part of the reviewed corpus. Each states its unproved hypothesis explicitly; none advances #249, and they are listed because a reader deciding whether to spend time here should see the sufficient conditions in the exact form Lean checks them.

Strict prime-orbit escape. The producer asks for cofinally many primes p at which the first tail-orbit exponential has real part strictly below $9/10$. Granting it, a positive adaptive truncation budget follows, and then S is irrational. The point of interest is the constant: the legacy route required a $4/5$ gap, this one requires only $9/10$, so the sufficient condition is strictly weaker while the conclusion is unchanged. Weakening a hypothesis you still cannot verify is progress on the *statement*, not on the problem, and the producer remains unproved. Section 7 states the sharper form in which this hypothesis is now available.

Finite Euler-sieve algebra. Elementary identities behind the squared-Möbius approximation to the totient generating function; no transcendence input is claimed. The local Euler factors are exact squares at $s = 1$ and $s = 2$, and on the divisor-sum prime-power row the first difference is $p-1$ while the second difference is $p^{e+1}(p-1)$. The second difference is the content: applying $\mu * \mu$ converts the divisor-sum row into the prime-power totient row exactly, at every finite stage. These are one-line facts; they are recorded because the approximation they license is otherwise easy to assert without checking.

Prime-ray cyclotomic curvature. A producer/consumer interface with two named obligations. Its curvature functional is the checkerboard on a two-anchor rectangle, which annihilates every separable background $u(x) + v(y)$ and is the unique such 2×2 annihilator up to scale. That uniqueness is worth stating: the operator is not a convenient choice among many, it is the only one, which is what makes the resulting obstruction coordinate-free. On top of it sit a squarefree divisor-layer cancellation and a centred-completion step for residue classes modulo an even modulus. The two obligations are eventual clean cyclotomic layers on a prime ray and an exact-order witness at every rational prime divisor of a layer; together they give escape from every prescribed finite prime support. Neither obligation is proved here, and both polynomial resultant realisability and Archimedean growth remain upstream of both. Obligation (b) is a primitive-divisor question in classical shape: it asks that every prime divisor of a layer $C(mq)$ have multiplicative order at most d relative to the ray index mq. One line, offered as prose and not as a checked statement: (b) alone already forces the escape by size, since $p \mid C(mq)$ gives $mq \le p^d - 1$, so for a finite set S of primes no member of S divides $C(mq)$ once $q > \max_{p \in S} p^d / m$; the role of (a) is then to keep the conclusion from being vacuous, since $C(mq) = 1$ would satisfy it trivially. Whether (b) is known, in either direction, for cyclic resultants of an integer polynomial is the question we would put to a specialist first.

7 The exact frontier

The conditional part of this work rests on one obligation, and it is worth stating in the sharpest form the corpus now supports rather than the form an earlier draft used.

The obligation. Write $H_t = \operatorname{lcm}(1, \ldots, t)$ (definition) and let $R_N = \sum_{m \ge 1} \varphi(N+m)/2^m$ be the fractional layer of $2^N S$ (definition). *Wanted:* for every t_0 there is a $t \ge t_0$ with $R_{2H_t} - R_{H_t} \notin \mathbb{Z}$. By R8 that supply is not merely sufficient; it is equivalent to #249. Nothing short of a cofinal statement is therefore of use, and no finite band approximates one. The converse direction is sharper than it looks: one pair $h > 0, N$ with $R_{N+h} - R_N \in \mathbb{Z}$ already makes S rational (checked).

The certificate is decidable at each t and has been kernel-checked to fire at the 28 scales (listed) lying between $t = 1$ and $t = 64$ (checked). Later work in the repository extends the checked band to every $t \le 82$; the module proving it postdates this note's pinned snapshot and so is not linked here, and the claim registry records the wider band. The distinction does not matter for the obligation: a band bounded at $t = 64$ contributes nothing at $t = 65$, and a band bounded at $t = 82$ contributes nothing at $t = 83$. That is the whole difficulty, and lengthening the list is not offered as progress towards it.

An analytic route, with the budgets made explicit. The most concrete approach the corpus supplies replaces the qualitative prime-orbit gap of Section 6 by a decomposition with numbers in it. A constant-saving first-harmonic bound on one dyadic block, or on any nonempty finite set of indices, already forces a certificate (checked). The block sum in question decomposes exactly into four named terms — a centred supplier-fibre correlation, a fibre-mean contribution, a bad-set contribution and a non-supplier contribution (checked) — and meeting four explicit numeric budgets on those terms cofinally already gives irrationality (checked). So the analytic obligation is not "prove a cancellation estimate" in the abstract: it is to beat four stated constants on a decomposition that is already exact. That is the form in which we would want it attacked, and it is the form in which a negative answer would also be informative.

What an answer would do. A proof of the supply closes #249, since the reformulation is an equivalence and the deduction is checked. A demonstration that the lcm diagonal is structurally the wrong ray, or that the four budgets cannot be met, retires a presentation without touching the problem, and we would record that here and drop the route rather than leave it listed as live. A single t with $R_{2H_t} - R_{H_t} \in \mathbb{Z}$ settles #249 in the opposite direction.

Not an ask. The constant 7.96×10^{34} of R3 is not an open item, and help with it is not wanted. It is the classical Farey mediant bound applied to a window with a free parameter K, it has order $2^{K/2}$, and raising it needs only more committed binary digits — a computation, not an idea.

8 What remains open, and what the registry does not yet own

8.1 Open

1. Irrationality of S (R10). No proof is claimed.

2. The certificate routes of R8 convert irrationality into the existence of an unbounded supply of finite integer certificates. Finite instances are checked — R3 is one, at $K = 240$ — and the unbounded supply is not. *An equivalence is not progress.*

3. The four pivot budgets of Section 7, and the prime-orbit gap they refine, are unproved.

The correlation source that fits the normalised totient most directly is Balasubramanian–Giri–Srivastav [5], not an off-the-shelf transfer of the Tao–Teräväinen engine. Write

$$g(n) = \frac{\varphi(n)}{n} = \sum_{d|n} \frac{\mu(d)}{d} = (f * 1)(n), \qquad f(d) = \frac{\mu(d)}{d} \in \mathcal{A}_1.$$

Theorem 2 of the arXiv version of [5] gives, uniformly for $|h| \leq x/2$, an explicit asymptotic for $\sum g(n)g(n-h)$ with error $O(\log^2 x)$; Remark 4 gives its Euler product, and the weighted partial-summation formula immediately following Corollary 8 permits $Q(n) = n(n-h)$, restoring the two linear factors needed to return from $g(n)g(n-h)$ to $\varphi(n)\varphi(n-h)$ with an explicit propagated error.

This corrects a tempting but inaccurate description in an earlier draft. Tao–Teräväinen's quantitative correlation theorem ([4], Theorem 10) assumes either quantitative equidistribution together with an exact small-prime condition, or non-pretentiousness. Here $g(p) = 1 - 1/p$, so the stated small-prime condition does not hold, while $\sum_p (1 - g(p))/p = \sum_p 1/p^2 < \infty$, so g is pretentious to the constant function. Their theorem therefore does not apply unchanged. The Balasubramanian–Giri–Srivastav theorem supplies the missing first- and second-moment input in the correct divisor-convolution class, but it does *not* supply the residue small-ball or phase anti-concentration estimate needed to turn those moments into a certificate. That modulo-2^L step is the precise remaining analytic gap; no irrationality conclusion is drawn from the correlation asymptotic alone.

8.2 Unregistered results

Two of the items above are checked propositions that no row of the claim registry currently owns: the three declarations of R1, and the five of R5; the supply implication of R9 is likewise unowned, and R3 cites three declarations of which the registry's row owns one. Under this project's own division of authority the registry, not a manuscript, owns public status, so the honest tag on those items is that they are kernel-checked and not yet registered. We print that rather than borrowing a neighbouring row's status. The gap is a defect in the record, not in the proofs, and closing it is a review action rather than a mathematical one.

Statements and declarations

This manuscript is authored exposition, not proof authority. The linked Lean snapshot is authoritative only for its exact propositions, and kernel checking establishes that a proposition was proved, not that it is interesting, novel, or sufficient. The consequence drawn in the last sentence of R1, the deduction in R6 that an unbounded weight is not eventually periodic, the mechanism sentence in Section 5, and the size argument in Section 6 are one-line arguments in the prose, marked as such; the first three are drawn from checked statements, and the last is drawn from two hypotheses that are not proved. Results attributed to Allouche and Shallit, to Coons, to Bell and Smertnig, to Luca and Tachiya, and to Tao and Teräväinen are cited from the literature and are not formalised here; the assessment that no prior source gives an explicit basis, an exact finite-level dimension formula, or a relation normal form for the totient 2-kernel is the outcome of a literature search and is not a proof of non-existence. Erdős Problem #249 remains open.

References

[1] J.-P. Allouche and J. Shallit, *The ring of k-regular sequences*, Theoret. Comput. Sci. 98 (1992), 163–197, DOI. Definitions 1.1 and 2.1 fix the k-kernel and k-regularity.

[2] M. Coons, *(Non)Automaticity of number theoretic functions*, J. Théor. Nombres Bordeaux 22 (2010), no. 2, 339–352; arXiv:0810.3709. Theorem 3.3: φ is not k-regular for any $k \geq 2$.

[3] J. Bell and D. Smertnig, *Mahler series with multiplicative coefficient sequences*, 2026, arXiv:2603.23456. The totient generating series is not k-Mahler for any $k \geq 2$, a consequence of their Theorem 1.3 recorded in the introduction.

[4] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (submitted December 2025, revised April 2026). Theorem 1.3 proves irrationality of $\sum_{n\geq1} \omega(n)/2^n$ at base 2; the extension to every integer base and the Ω analogue are stated as remarks, with the modifications left to the reader.

[5] R. Balasubramanian, S. Giri and K. R. Srivastav, *On correlations of certain multiplicative functions*, J. Number Theory 174 (2017), 221–238, DOI; arXiv:1511.02221. The arXiv version's Theorem 2 and Remark 4 give the uniform shifted divisor-convolution correlation and Euler product used above.

[6] F. Luca and Y. Tachiya, *Irrationality of Lambert series associated with a periodic sequence*, Int. J. Number Theory 10 (2014), no. 3, 623–636, DOI.

[7] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, 1980, p. 61.

[8] T. Bloom, *Erdős Problems*, `https://www.erdosproblems.com/249`.